

pcap trace files will be saved in the directory `ns/ns-allinone-3.22/ns-3.22`.

The pcap trace files will not give you any information if opened in a text editor. To elicit information, you have to process these files with a packet-analyzing utility like tcpdump or Wireshark. First, let us see how to use tcpdump to extract information from the pcap trace files. The utility tcpdump is a packet analyser that runs in the command line. It has the ability to generate pcap trace files as well as analyse it. Since pcap trace files are already generated by ns-3, tcpdump is used only for analysis. Open a terminal in the directory `ns/ns-allinone-3.22/ns-3.22` and execute the following command:

```
tcpdump -nn -tt -r pcapsim-0-0.pcap
```

This command will process the pcap trace file `pcapsim-0-0.pcap`. Figure 2 shows the output of applying tcpdump on the two pcap trace files `pcapsim-0-0.pcap` and `pcapsim-1-0.pcap`.

The option `-tt` instructs tcpdump to print unformatted time on each line of the tcpdump output. The option `-r` instructs tcpdump to read and process an input file. As mentioned earlier, tcpdump has the ability to observe a network and produce a pcap data file. But here, pcap trace files are already available and hence the option `-r` is used. The option `-nn` instructs tcpdump not to convert the IP address to the hostname or the port number to the port name. For example, the IP address `64.233.166.94` will be converted to `google.co.in` and port number `80` will be converted to `HTTP` (remember, port number `80` is the well known port for HTTP) if the option `-nn` is not used in tcpdump. If you use the option `-n` instead of `-nn`, then the port number will be converted to the corresponding port name. Assume that UDP echo server port number was `80` in the script `pcapsim.cc`, then the command `'tcpdump -n -r -tt pcapsim-1-0.pcap'` will give the following output on the terminal:

```
3.005433 IP 192.168.100.1.49153 > 192.168.100.2.http: UDP,
length 512

3.005433 IP 192.168.100.2.http > 192.168.100.1.49153: UDP,
length 512

6.005433 IP 192.168.100.1.49153 > 192.168.100.2.http: UDP,
length 512

6.005433 IP 192.168.100.2.http > 192.168.100.1.49153: UDP,
length 512
```

As you can see, port number `80` is replaced by the port

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.100.1	192.168.100.2	UDP	542	Source port: 49153, Destination port: cisco-sctp
2	0.001000	192.168.100.2	192.168.100.1	UDP	542	Source port: cisco-sctp, Destination port: 49153
3	0.000000	192.168.100.1	192.168.100.2	UDP	542	Source port: 49153, Destination port: cisco-sctp
4	0.001000	192.168.100.2	192.168.100.1	UDP	542	Source port: cisco-sctp, Destination port: 49153

Figure 3: Wireshark window showing pcap trace file

name `http`. Since tcpdump is a very powerful utility, a lot of information can be extracted from pcap trace files by using the different options of tcpdump. But many new users of Linux and ns-3 feel a general aversion towards tcpdump because it is a command line utility. If you, too, feel the same way about command line utilities then Wireshark is the solution to your problems.

Wireshark is a packet analysis utility with a graphical front-end (GUI interface). You can open the pcap trace files in Wireshark and view the results on a window. Figure 3 shows the Wireshark window showing the pcap trace file `pcapsim-0-0.pcap`. In the figure, you can observe the fact that registered port number `2000` is identified with the port name `cisco-sctp` by Wireshark. A lot of options for filtering and analysis are available with Wireshark, which can be used to extract lots and lots of useful information from pcap trace files. A detailed discussion about Wireshark is beyond the scope of this tutorial but you can find a number of Wireshark related documents at www.wireshark.org/docs/.

A few words about the ns-3 source code

We have used many built-in classes of ns-3 to simulate different networks. What about the source code of these built-in classes? We have already discussed Doxygen and how it helps to maintain the documentation of ns-3. You can find the ns-3 documentation at www.nsnam.org/doxygen/. You can get details regarding ns-3 modules, namespaces, classes, etc, from there.

But what is the importance of ns-3 source code? We know for a fact that many network protocols and network related concepts can be simulated in ns-3. But what if your research project involves the creation of a new network protocol? Then you need to alter the source code of ns-3 to incorporate your new protocol into the existing structure of ns-3. So it is absolutely essential to go through the source code of ns-3. But remember, there are hundreds of thousands of lines of code in ns-3 and it is not humanly possible to go through all of it alone. So you must identify the ns-3 source files relevant to your research, have a deeper understanding of the code in them, and finally modify them. Once you have made changes to the ns-3 source code, you should rebuild the entire source code so that the modified code will have its effect on future simulations. If you have successfully altered the source code of ns-3 to create a new protocol, please share it with